

Skill Arena - Phase 4 Document 3

Anti-Cheat Engine, Gameplay Validation, Replay Verification & Match Integrity Architecture

Document Purpose

Define how gameplay is validated, cheating is prevented, replays are verified and match integrity is maintained.

Business Objective

Ensure every victory, defeat, reward and ranking outcome is earned through legitimate gameplay.

Security Philosophy

The server is the authority for all gameplay outcomes. The client is only an input and display layer.

Anti-Cheat Engine

Central system responsible for detecting, preventing and investigating gameplay manipulation.

Gameplay Validation Engine

Validates every move, timer event, life deduction, completion event and reward calculation server-side.

Server Authoritative Design

The server determines challenge state, progress, completion and rewards.

Client Trust Level

Zero. No gameplay result generated by the client is trusted.

Movement Validation

All player movement requests validated against challenge rules and maze structure.

Timer Validation

Challenge timers controlled exclusively by trusted server infrastructure.

Life System Validation

Life deductions calculated and recorded server-side.

Completion Validation

Challenge completion requires server verification before rewards are issued.

Replay Generation Engine

Creates immutable replay records for all live matches and house challenges.

Replay Verification

Each replay receives verification identifiers and integrity validation records.

Replay Storage

Replay metadata and event history retained for audits, disputes and investigations.

Match Integrity Framework

Protects competitive fairness across PvP, tournaments and house challenges.

Copy Prevention Controls

Opponent routes hidden, delayed previews and anti-observation safeguards enforced.

Challenge Randomization

Challenge generation designed to reduce predictability and replay exploitation.

Pattern Detection

Monitor for unusual completion rates, impossible actions and suspicious behaviour.

Bot Detection

Identify automation, scripted inputs and non-human gameplay patterns.

Collusion Monitoring

Detect coordinated behaviour designed to manipulate rankings or rewards.

Replay Investigation Tools

Investigators can review historical gameplay and integrity events.

Tournament Integrity Controls

Additional validation applied to championship and tournament matches.

Security Event Logging

Gameplay events, validation failures and integrity alerts permanently recorded.

Incident Response

Automated and manual workflows for suspected cheating incidents.

Audit Requirements

Every gameplay validation and anti-cheat action recorded for future review.

Performance Requirements

Validation and anti-cheat systems must operate in real-time without degrading gameplay.

Acceptance Criteria

Platform can validate gameplay, detect cheating and defend competitive integrity at scale.

